

Exploring Computational Thinking Practices and Gestures in the Context of Matrix Math

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ABSTRACT

Computational thinking (CT) is a unique skill set that can provide advantages not only in the field of computer science (CS) but also in other disciplines. There is potential benefit in incorporating CT practices in other subjects so students from interdisciplinary backgrounds can also gain CT skills and apply them in their respective disciplines. This study takes a look at how students from interdisciplinary backgrounds interact using gestures and use CT to understand matrix math concepts in a group setting. A mixed-methods analysis is used to gain insight into how students' practice CT. Data collected from pre- and post- assessments measures students' mastery level in matrix math concepts. Video data collected during the group activity shows students' learning process. The results of this study provide a better perspective on how other disciplines are able to utilize CT and how CT learning can be implicitly taught in subjects other than CS.

CCS CONCEPTS

• **Applied computing** → **Interactive learning environments.**

KEYWORDS

computational thinking, mathematics, gestures, interaction

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1 INTRODUCTION

Computational thinking (CT) is a concept that has been traditionally associated with the field of computer science (CS). However, in the present day, CT is a skill that is necessary for many fields outside of CS. In order to encourage CT practices in interdisciplinary studies, we need to first understand how students from interdisciplinary fields interact with and use CT practices. This will help inform future designs of collaborative, technology-supported learning in higher education.

This work aims to answer the following research question: *How do students with interdisciplinary backgrounds use computational*

thinking practices in the context of learning matrix math with robotics? The objectives of this analysis are to understand how students practice CT in this context and to identify what tools they use to communicate while employing CT practices. The study of matrices incorporates concepts of linear algebra and discrete mathematics. Matrix math was chosen as the medium of instruction due to its versatile applications in interdisciplinary fields. Additionally, mathematics is frequently one of the most commonly identified subjects that students find difficult. CT can make it easier to learn matrix math as there are many discrete mathematical concepts that can be taught from the perspective of CT and vice versa [10].

2 BACKGROUND

Computation thinking is a multifaceted concept that has multiple definitions and applications. In general terms, it can be thought of as the thought process in which computer scientists think. Wing (2008) introduced this concept as “taking an approach to solving problems, designing systems and understanding human behavior that draws on concepts fundamental to computing” [9]. Other varying definitions have been proposed in recent years, with varying views on the breadth of applicability of CT. Some consider CT to be a concept closely limited to computers and programming. Others view CT as a broader idea that includes and has more applications beyond computers [4]. Brennan and Resnick (2012) introduced a definition for CT that includes three dimensions: computational concepts, practices, and perspectives. This framework can be adapted to apply to diverse learning circumstances.

As CT becomes an increasingly essential skill, various education institutions have begun incorporating CT standards into their curriculum. The Computer Science Teachers Association revised their K-12 Computer Science Standards in 2017 to include CT concepts and practices [1]. As much as it is necessary to begin introducing CT to students in K-12, it is also essential to take a look at different methods for teaching CT in higher education. Due to the nature of CT being applicable to interdisciplinary fields, it is necessary to promote CT learning beyond the computer science discipline.

In this study, we take a look at one approach that allows students from interdisciplinary backgrounds to practice CT while learning matrix math. This approach incorporates the use of physical materials, in the form of a robot arm, which are found to have a strong effect on mathematics learning [6]. The integration of robotics in the design of this study helps us explore the possibilities of CT in interdisciplinary fields in higher education.

3 METHODS

This pilot study is part of a larger project that aims to explore how computer science curriculum can be reconstructed to teach CT with

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Table 1: Pre and Post Assessment Scores on Learning Objectives

Learning Objective (LO)	Average Pre	Average Post	Gain
LO 1: Vector Operations	0.25	0.42	0.17
LO 2: Spatial Reasoning / Mental Rotation	0.55	1.66	1.11
LO 3: Matrix Operations	0.11	0.67	0.56
LO 4: Robotics	0.11	0.78	0.67

the use of tangible fiber crafts. A robot arm serves as a physical material that can potentially enhance the interdisciplinarity of this activity in this pilot [6, 7]. This pilot was conducted at a four-year public university in a graduate-level four-unit elective course called Arts, Robotics, and Engineering. One of the units of this course focused on kinematics, during which students completed a robot arm group activity designed to teach matrix math.

3.1 Participants and Setting

This study consisted of 10 participants in four groups of two to three people each. This analysis focuses on a three-person group of students who were pursuing a Doctorate of Philosophy in Education at the time of the study (i.e., Group 3). One of the participants also had a background in Computer Science.

The activity was conducted in-person with the lead instructor, Rachel, joining remotely via Zoom. The groups were provided a worksheet with the steps of the activity and helpful hints. The pace of the activity was set by the lead instructor as the groups relied on and followed her instructions.

3.2 Activity

The participants watched a one hour long pre-recorded lecture before the activity to learn the fundamentals of kinematics. Each group was provided with a commercial robot arm kit called LeArm. The robot included 6 servo motors (servos) that allowed it to bend each of its 6 joints. The software that accompanied the robot allowed users to rotate each servo to the desired angle by moving a slider right or left. The arm also included a claw as the end effector. The robot was set in front of a board that had a coordinate plane on it. The initial state of the arm had each servo at 0° so that the arm was pointing straight up.

One member per group was responsible for assembling the robot arm and bringing it to class for the activity. During the activity, students had a chance to play with the robot arm. They discussed among themselves and then rotated servos 3, 4, and 5 by various degrees. They found the new coordinate point of the end effector based on the coordinate plane that was provided for them.

The group was then guided by the instructor to use matrix math to calculate the final position of the end effector based on the measurements of the robot arm, such as the lengths of its segments and the angles of rotation for each servo. They began by creating a diagram for their robot arm with 3 points representing the location of the servos that had been rotated. Next they drew in the corresponding frames and angles for each point and discussed what the axis of rotation was. They used this information to express the rotation of each servo with matrices and calculated the final coordinates of the end effector after all the rotations. The activity

was concluded with a discussion on the possible errors and reasons that led to the math coordinate being slightly different from the real life coordinate. The students shared their thoughts at the end of the activity and took part in a reflective interview.

3.3 Data Sources

The principal data sources were video data and pre/post assessments. Videos recorded from multiple angles during the activity documented participants' body language, communication, and collaboration. Approximately 7 hours and 36 minutes of video footage was collected from all 4 groups. There was approximately 1 hour and 48 minutes of video data for the focal group described above and I analyzed roughly 21 minutes of this data. I selected video data for analysis based on the meaningful discussions taking place that were related to the matrix math Learning Objectives (LOs):

- LO 1: Vector Operations
- LO 2: Spatial Reasoning / Mental Rotation
- LO 3: Matrix Operations
- LO 4: Robotics

Another critical data source was the written assessment that participants took before and after the activity. This assessment included conceptual and mathematical questions to measure mastery level in the 4 pre-determined Learning Objectives (Table 1). Altogether, 10 pre-assessments and 9 post-assessments were collected and I analyzed the 3 pre-assessments and 3 post-assessments of Group 3.

3.4 Data Analysis

I conducted a qualitative video analysis by transcribing meaningful segments that were related to the Learning Objectives [5]. I adapted Brennan and Resnick's framework for studying the development of computational thinking as my coding scheme. The CT practices that I coded for were Incremental & Iterative (I), Testing & Debugging (T), and Abstracting & Modularizing (A) [2]. I also analyzed the gestures used by students in the activity when they were trying to communicate their thoughts and illustrate the rotation of the robot arm.

4 FINDINGS

4.1 Quantitative

Improvement was seen in each LO when comparing the pre and post assessments for Group 3. The mean score for Group 3 on the pre-assessment was 3.33 and on the post-assessment was 11.00. LO 1 had a gain of 0.17 in average score, LO 2 had a gain of 1.11, LO 3 had a gain of 0.56, and LO 4 had a gain of 0.67.

The improvement seen in these LOs was directly linked to the discussions that took place during the activity as can be seen in the video data. Some of the points of confusion that the students encountered in the activity were related to, if not the exact, questions on the assessments. For example, there were multiple questions that the students did not answer on the pre-assessment. During the activity, students discussed the concepts related to those questions and then were able to answer them on the post-assessment.

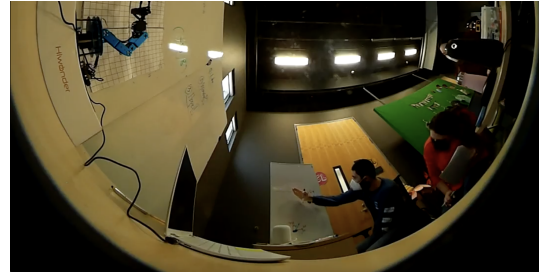
The greatest gain achieved was in Spatial Reasoning / Mental Rotation (LO 2) which increased from an average score of 0.55 in the pre-assessment to an average score of 1.66 in the post-assessment. The video data provides evidence that this gain is correlated with the use of gestures and visual aids [6–8]. A majority of the gestures used by participants occurred when discussing LO 2. Additionally, discussion of LO 2 was often accompanied by CT practices such as Testing & Debugging. When discussing LO 2 students interactivity, in the form of CT practices and gestures, greatly increased. This increase in interactivity suggests that there was enhanced engagement and learning taking place.

4.2 Qualitative

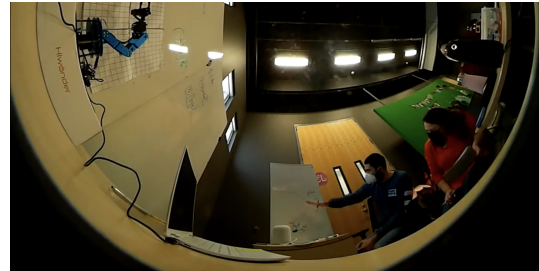
Video data showed abundant evidence of CT practices and gestures being used by the participants during the activity. Vignette 1 (Table 2) was selected to highlight how students interacted with the diagram and the gestures they used to communicate their ideas and confusion. Given that the students were not able to draw on the robot arm, it was necessary to have another means of representing the robot arm. In this case, the students used a whiteboard for their drawing which they could then manipulate and write notes on. Interacting with the manipulator diagram of the robot arm that they had drawn can be considered equally as valuable as interacting with the robot arm directly [6–8].

Throughout the activity, students used different gestures with their hands and arms to model the robot arm's movement and rotations. For example, in Vignette 1 (Table 2, Lines 8–16), Josh (all names are pseudonyms) uses his hand to communicate the motion that he is talking about. First he holds his hand at a 45° angle parallel to the robot arm to show the angle that the arm has been rotated (Fig. 1(a)). Then he holds his hand horizontally at joint A and slides it up along the arm to joint B to demonstrate the plane that he is picturing in his mind (Fig. 1(b)). Similarly, Paige has some difficulty explaining her point of confusion (Table 2, Lines 19–21). She uses gestures to assist her in describing her idea. She holds her hand up straight and then moves it counterclockwise to illustrate the motion of the angle that she is talking about (Fig. 1(c)).

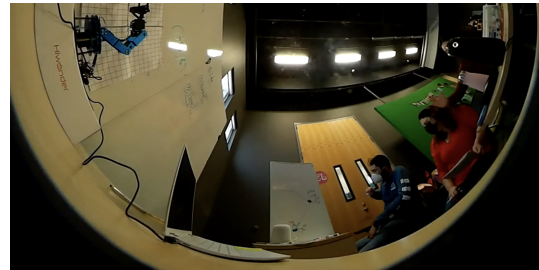
In this vignette, the group experienced some confusion with LOs 1 and 2. Paige noticed there was something incorrect about the drawing Josh had made. Although she did not identify the correct mistake, she was able to stimulate a discussion in the group which ultimately led to Lauren and Josh figuring out what was wrong with the drawing. From the point where Paige pointed out her concern to the point where the group resolved the issue, they greatly utilized the CT practice of Testing & Debugging. They were able to use Testing & Debugging to pinpoint their misunderstanding and revise their diagram without needing the instructor's help.



(a) Josh uses his hand to imitate the position of the robot arm segment.



(b) Josh uses his hand to demonstrate the plane that vector A_y is parallel to.



(c) Paige uses her hand to show that the robot arm rotated in the counterclockwise direction.



(d) Lauren uses her hand to mimic the motion of the servos moving.

Figure 1: Gestures employed in activity to model robot arm's movement.

Vignette 2 was selected to illustrate how learning matrix math and discussing LO 2 and LO 3 helps students strengthen the CT practice of Abstracting & Modularizing. The collaborative environment that was created in this activity stimulated students to think deeper

Table 2: Vignette 1: Writing in A and B frames

Line	Verbal	Gestures	CT	LO
1 2 3 4 5	Rachel: Ok so now you're going to move along the next joint into the second joint and that joint is going to be rotating relative to A and it rotated some amount and you'll probably want to write a B frame that represents that rotation relative to A.	Rachel holds her arm at a $\sim 45^\circ$ angle and moves it from the bottom right of the screen to the top left to illustrate what she means by moving along the arm to the next joint.		4
6 7	Rachel: So in this second joint can you also do the same thing, but with the A and the B frame?			1
8 9 10 11	Josh: Ok this makes a lot of sense cause this is why this angle's like all the way, ok so the B would normally go up this way, but because we've got it rotated so far, ok that makes sense.	Josh holds his hand in front of the board along the robot arm when referring to the rotation of that joint to show the angle that has been rotated.	I, A	2
12 13 14 15 16	Josh: And then, oh so like our A frame mimics this like if A was represented as a plane, it would be following the same way so our A continues to go this way so this is our A_y and then our A_x would be parallel to the one that's down there so it's like that.	To show the plane, Josh holds his hand horizontally at joint A and slides it along the diagram to joint B.	I, A	1
17 18	Rachel: How do we feel about that rotation in the frames?			2
19 20 21	Paige: Because it's not following the same, kind of the same angle as the, would our A be more kind of like counterclockwise a little bit?	Paige uses a pen to point to the arm segment that represents B to C. She also holds her hand up straight and then moves it counterclockwise to illustrate the movement of the arm.		2
22 23 24	Paige: Looks like a 45 degree angle from the slope of the arm following the joint, would it be, I mean I don't know.	Paige uses pen again to point to the arm segment that she is referring to.	T	2
25 26 27 28	Josh: My assumption would be that our B is what goes, like I think our A might be okay, because our A is meant to align with what's down here, but our B would be the one that's changed because	Josh uses marker to point to the joints so the group knows which joint he is talking about.	T	1, 2
29 30 31 32	Lauren: Yeah I think what you did is right. The A_y continues and the A_x goes in parallel and then we need to put the B. Thank you for asking that because it took me thinking more.		T	1, 2
33 34	Josh: So our B is this way, so our B_x might, would our B_x be the same as this?	Josh draws vectors on the diagram.	T	1, 2
35 36	Paige: Yeah I feel like it's perpendicular as long as it's following that arm right.		T	1, 2
37 38 39 40 41	Josh: Our A_x and A_y should be the same here, but we know our B_y would be the axis of the arm so our B_x would just be perpendicular to that at a 90 degree angle so it seems like it's the same as our A_y , but that might just be a coincidence.	Josh uses marker to point to which part of the diagram he's talking about as he explains. He also uses the marker to draw in the air the 90degree angle he is referring to.	A	1, 2

about how the robot arm itself worked. Given that the majority of this group does not have a background in Computer Science, it is evident that being able to make such connections between the mathematics and kinematics in this activity using CT practices promotes an interdisciplinary understanding. The Abstracting &

Modularizing CT practice enables students to develop interdisciplinary skills as they understand and propose connections between mathematics and kinematics.

When trying to understand abstract concepts such as how the robot moved or why the robot was rotating around the z-axis,

Table 3: Vignette 2: What are we rotating around?

Line	Verbal	Gestures	CT	LO
1 2 3 4 5 6	Rachel: So the first thing I like to do is what rotation matrices do we have here? And what axis are they rotating around? There's 3 rotations so there's maybe 3 rotation matrices that you can write and what are these rotations rotating around, what axis?			2, 3
7 8 9 10	Josh: I have a guess but I don't think it's very sound. I think it's the Z matrix because that's what we had to look at in the lecture. I don't think I understand why it's the Z matrix.		T	2
11 12	Rachel: Let's talk about why it's the Z matrix. Anyone have any ideas?	Paige shakes her head.		2
13 14	Josh: Well the Z axis is the one that points out at us, this way.	Josh holds marker against the whiteboard to represent z axis	I	2
15	Lauren: That's where the motor is.	Lauren holds a pen between her thumb and index finger to represent the z axis. She then rotates her wrist back and forth to show the motion of the servo and how her hand moves around the pen.	T	2, 4
16 17	Lauren: Cause that's where the motor is, that goes like this, it has that motion.		A	2, 4
18 19	Rachel: Yeah, if you look, the motor's kind of rotating along that pole.	Same gesture as before	A	2, 4
20	Lauren: That's how the servo moves.	Josh and Paige nod.	A	2, 4

students employed various gestures to model the movement of the robot arm. These gestures involved their bodies (e.g., hands and arms) as well as physical objects (e.g., pens and markers). One instance of this can be seen on Lines 13–14 of Table 3 as Josh holds a marker perpendicular to the whiteboard to replicate the z-axis. This action corroborates the idea that incorporating hands-on elements to math learning is beneficial in building a deeper understanding of the subject [6]. In this case, the use of physical objects makes it easier for the participants to visualize their ideas.

Another example of the gestures involved in this excerpt is seen on Line 15 (Table 3) when Lauren uses her hand and a pen to imitate the movement of the servo. She uses a pen to represent the z-axis and holds it between her thumb and index finger. Then she rotates her wrist back and forth to show the motion of the servo (Fig. 1(d)). This makes it easier for the other group members to visualize the z-axis (pen) staying fixed in its position and the robot arm (hand) rotating around it.

5 DISCUSSION

In the vignettes presented above, we can see how the interactive group activity stimulated students' creativity to take advantage of the resources around them. Throughout the activity students continued to actively use the tools around them to help them communicate. This prolonged engagement is likely due to the physical materials available to the students. Across all the groups, persons responsible for assembling the robot arm expressed their feelings of accomplishment and pride in successfully building the robot

arm. This reinforces the need for more hands-on activities in higher education and particularly in interdisciplinary fields that involve CT.

Since these students did not have a strong background in Computer Science, they expressed some frustration with working with the new terminology in this lesson. They also expressed their appreciation for having the whiteboard and diagram as being able to visually see their work through the whole activity made it easier to remember the terminology. And being able to check themselves that they are using the terminology correctly provided reassurance and confidence. Evidence of this can be seen when comparing Group 3's answers on the pre and post assessments. On the pre-assessments, students did not attempt many of the questions if they didn't fully understand them. However, on the post-assessments, they attempted nearly all the questions. Even if they did not have the entire correct answer, it was clear that they were more confident in their understanding of the learning objectives. For example, Question 1a on the assessment was related to LO 1 and asked the students to draw two given vectors on a coordinate plane. On the pre-assessment, Josh did not know what a 'vector' was so he drew coordinate points instead. After using this terminology in the activity, Josh was able to correctly draw the vectors on the post-assessment. This suggests the importance of students having the opportunity to interact with and use terminology in order to properly grasp different concepts.

Breaking down the questions on the assessment also showed that there was more improvement in conceptual understanding as

compared to mathematical computing. This is likely due to all of the discussion and collaboration that took place during the activity. Understanding and remembering the steps needed for a calculation typically takes some practice, particularly with intricate topics such as matrix multiplication. Since the students only set up these calculations during the activity and did not solve them by hand, it is expected that there was no significant improvement in the questions related to calculations.

6 CONCLUSION

The results of this study provide insight on how students from interdisciplinary backgrounds use CT practices and gestures when learning matrix math with robotics. Given the extensive applications of CT in interdisciplinary fields, it is necessary to explore how CT can be taught in an interdisciplinary context [3]. In this study we saw how students used CT practices to communicate their thought processes and clarify their misunderstandings. During the activity, students also used various gestures to mimic the movement of the robot arm when explaining their ideas. This mimicry facilitated students' learning as it supplemented their understanding of matrix math and made it easier for them to understand the problem from the robotics perspective [8].

This study provided the design of one interdisciplinary activity that combined matrix mathematics, robotics, and CT. The hardware present in this study (i.e., robot arm) was beneficial for students' dimensional understanding. Future work should explore other designs for interdisciplinary activities that can promote CT in higher education. It is also worth considering a design that links software engineering and hardware to examine the potential of CT in such a context.

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